ry3_analysis

MATLAB scripts for calibrating and analysing experimental microscopy images using the RY3 molecular rotor. These are provided as an example of a molecular rotor calibration and analysis workflow, but may need substantial modification for use with other study systems and microscopes. Refer to the manuscript for for an explanation of the method and definition of terms.

Getting started

Everything can be downloaded from MATLAB Drive using this link:

https://drive.mathworks.com/sharing/1ea5c25f-4430-4d06-97f7-bbf91846fdc1

Users can download and use on their computers if they have MATLAB installed or they can upload it to a free [MATLAB Online] instance (they only need to create an account).

Dependencies

This code has been developed and tested in MATLAB 2022a and 2024b.

- MATLAB Image Processing Toolbox (available for free in MATLAB Online)
- MATLAB Curve Fitting Toolbox (available for free in MATLAB Online)
- Bio-Formats MATLAB Toolbox (bfmatlab, not required for this workshop)
- Binary image distance transform for variable aspect ratio (bwdistsc, included with codes)

Overview

0. Standardization

rc_0_standards.m

The raw and unmixed viscosity standard image files are loaded and the mean fluorescence intensities are converted to viscosity relative to the initial calibration curve. The ratio of the two standard viscosities is saved out for scaling the experimental image calibration in *rc_b_calib.m*.

1. Deconvolution

rc_a_deconv.m

For a single experimental sample, both the raw and unmixed files are loaded and the raw blue RY3 (I_B), unmixed red RY3 (I_R), and SYBR Green I channels are extracted. All channels are deconvolved with the point spread function of the microscope and the resulting image matrices and metadata are saved out as a single struct (labelled ‘a’).

2. Calibration

rc_b_bgselect.m (optional)

Background regions free of objects should be selected to calibrate the experimental images relative to background. This graphical user interface (GUI) allows rectangular regions to be drawn on each I_R image plane and saves the coordinates to the corresponding '_a.mat' file for use in *rc_b_calib.m*.

rc_b_calib.m

Loads the '_a.mat' file, applies spatial filters, and calibrates the image matrices to viscosity relative to background and scaled by the viscosity standards. Saves out relative viscosity and SYBR image matrices and calibration metadata to a single struct (labelled 'b').

viscView.m (optional)

A GUI used to simultaneously view individual planes of the red RY3, blue RY3, SYBR, and relative viscosity images saved in the '_a.mat' and '_b.mat' files.

3. Viscosity probabilities

rc_c_sybrignore.m (optional)

View the SYBR 'b' images next to thresholded images with labelled connected components (bacteria and other cell objects). This is used for testing SYBR thresholds and identifying objects to include or exclude from the *rc_c_halos.m* analysis.

rc_c_viscignore.m (optional)

View the viscosity 'b' images next to thresholded images with labelled connected components (mucus objects). Allows testing of viscosity thresholds and identifying objects to include or exclude from the *rc_c_halos.m* analysis.

rc_c_halos.m

Loads the '_b.mat' file, identifies bacterial objects, defines 'halo' regions around the bacteria, partitions mucus and non-mucus regions, and calculates the probability of viscosities in bacterial halos and regions. Saves out probability counts, relative viscosity image matrices for each partition, and associated metadata to a single struct (labelled 'c').

Example Usage

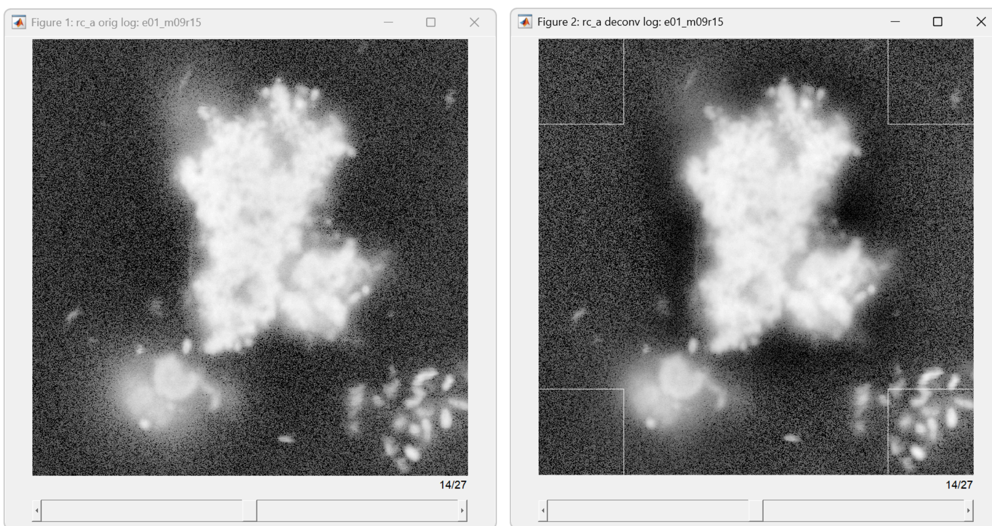
Download the sample file. Sample file description:

- **e01_m09r15_a.m**: A z-stack of a *Chaetoceros affinis* diatom aggregate imaged using optical configuration 15 during the 1st microscopy session. File size: 163 MB.

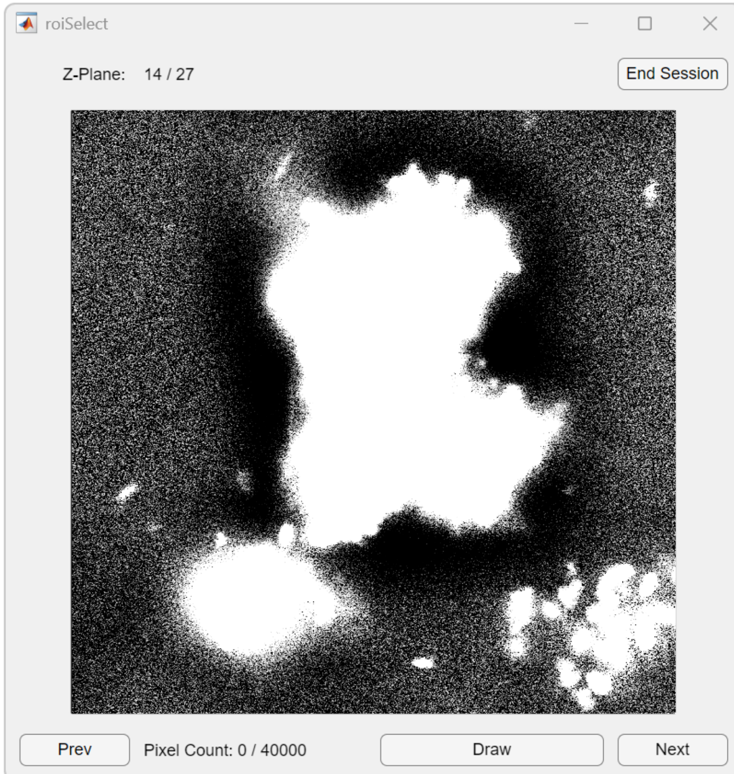
Place the code files in a directory named 'ry3_analysis' in your MATLAB directory.

The initial steps have been completed but are included here for reference:

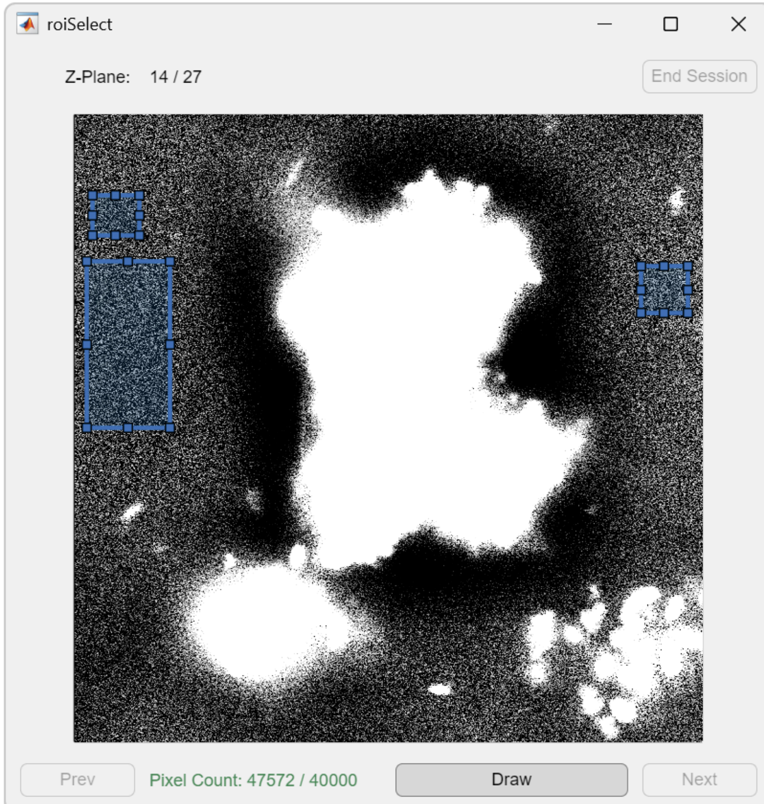
> You will first analyze the images of the viscosity standards for use in the calibration. Make sure the Bio-Formats toolbox is on your MATLAB search path (see Dependencies, above). Edit *rc_0_standards.m* and modify **fld** with the directory path to your sample images. The rest of the parameters are already set for use with these sample files. Running the script saves the ratio of the calibrated viscosity standards for this experiment to rstd.mat and displays their values.
>
> The next step is extracting relevant channels and deblurring them with the point spread function of the microscope. Edit *rc_a_deconv.m* and add the directory path to **fld** and the raw file name (e01_m09_r15.nd2) to **fnr**. Set **ztrunc**=7:33; to remove extra planes above and below the aggregate from the calibration. Running the script takes ~1 minute, saves the deconvolved SYBR, red RY3, and blue RY3 images to the '_a.mat' file, and displays the results of the deconvolution of the red RY3 channel:



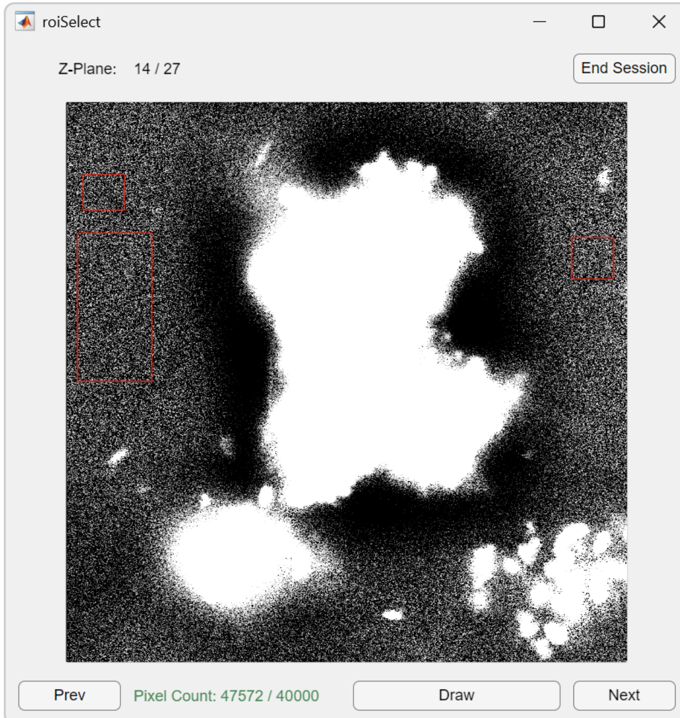
Calibrating the experimental images requires selection of background regions in every plane of the z-stack. These regions should be free of objects (e.g. aggregates and bacteria) and gradients of pixel intensities caused by optical artifacts or the deconvolution. You can select the background regions using the *rc_b_bgselect.m* GUI or enter them manually in *rc_b_calib.m*. To use the GUI, edit *rc_b_bgselect.m* and set **nm**='e01_m09r15':



The image is intentionally 'blown out' to make the background visible, but this can be changed by modifying the color scale limits **clm** in *rc_b_bgselect.m*. Toggle 'Draw' to begin selecting rectangular background regions (blue boxes):



Multiple regions can be selected, but they cannot be modified at this point. Select enough regions to satisfy the Pixel Count, and then untoggle ‘Draw’ to continue. Your selections are now displayed in red:

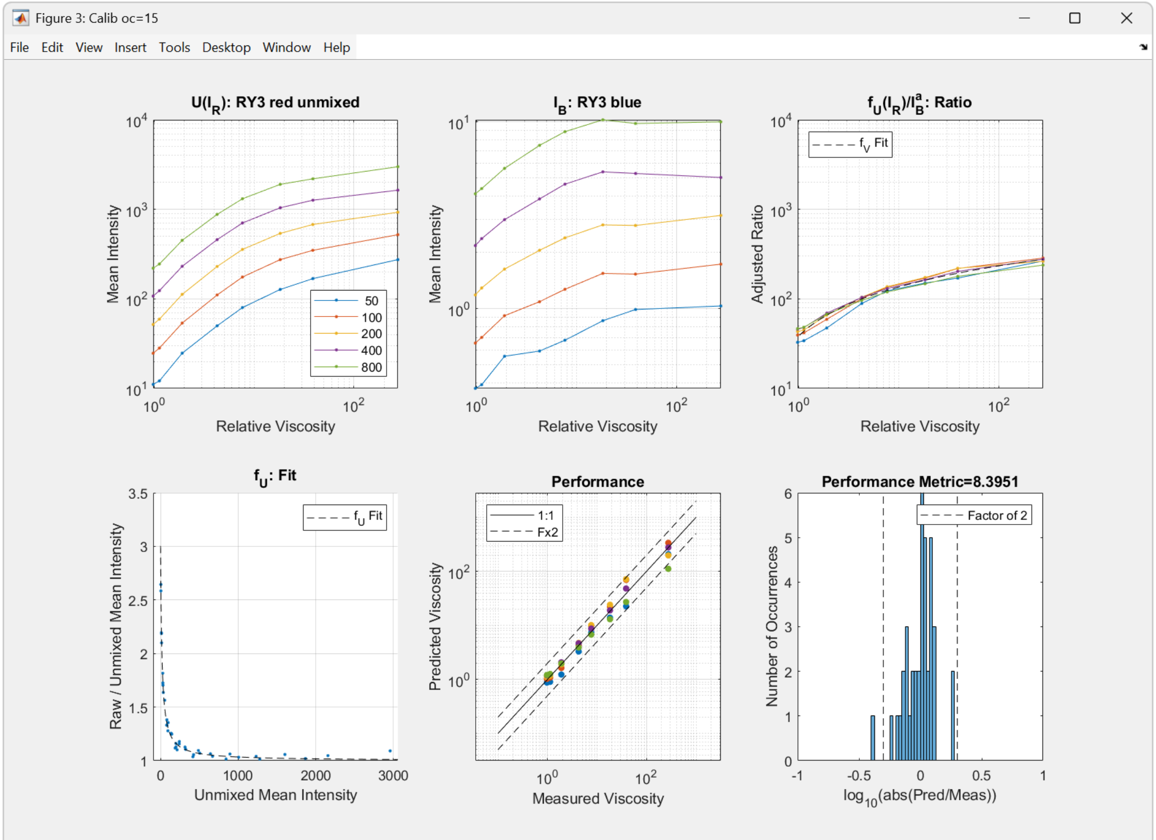


Click 'Next' or 'Prev' to move between planes. To modify previous selections, toggle 'Draw' and answer 'No' in the 'Add more ROIs?' dialog box. Untoggle 'Draw' to continue. After background regions have been selected in all planes, click 'End Session.' If **saveout**=true in *rc_b_bgselect.m*, the background region data will be saved to the '_a.mat' file.

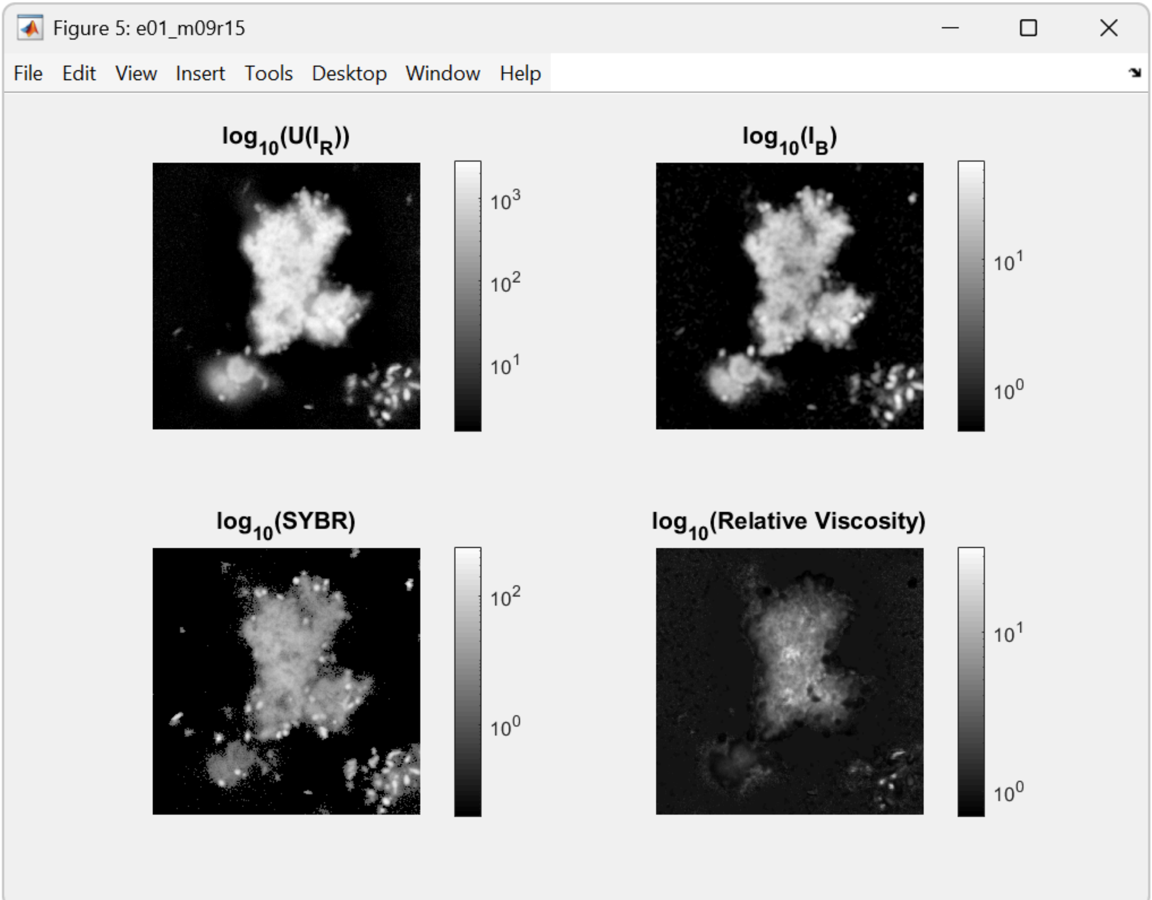
Next is the calibration of the red and blue RY3 channels to produce a relative viscosity image. Edit *rc_b_calib.m* and set **nm**='e01_m09r15'. If you selected backgrounds using the GUI, set **bgin**={}. If not, use:

```
bgin={  1:4,      1:200,     1:200;
        5:9,      824:1024,  1:200;
        10:13,    1:200,     824:1024;
        14,   160:560,   1:100; };
```

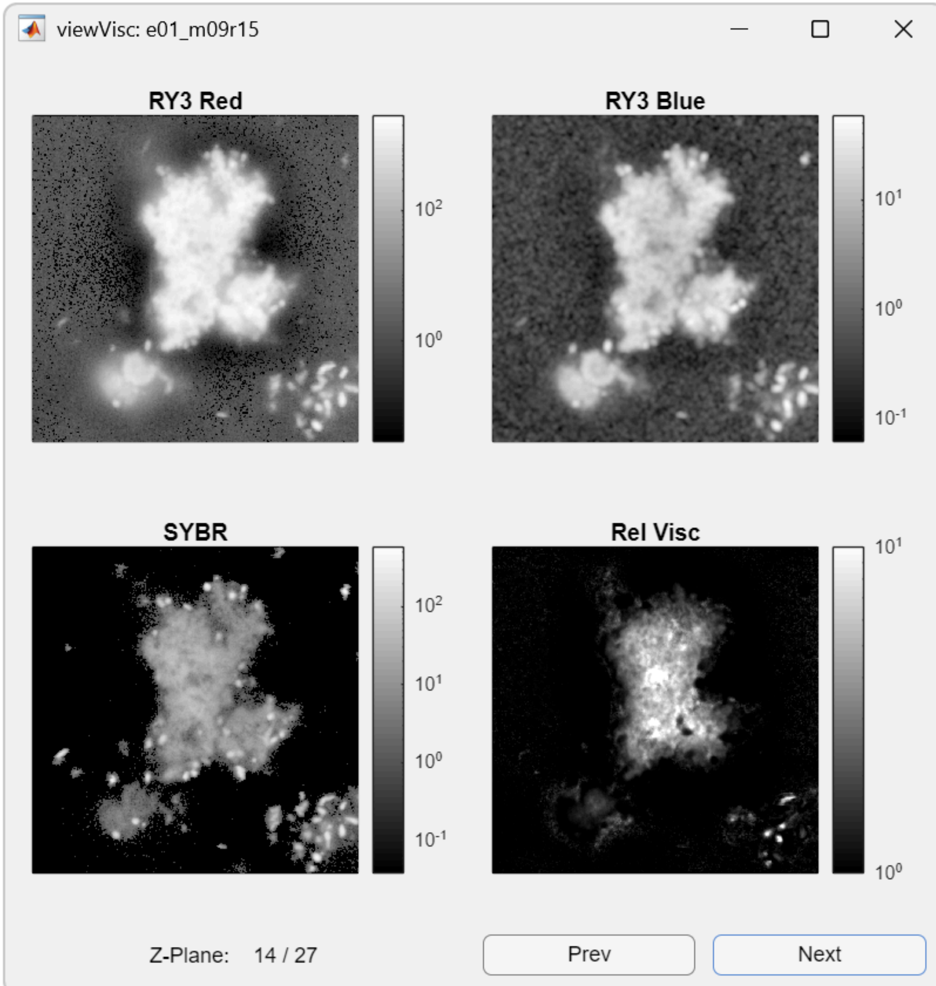
and leave all other parameters unchanged for this example. Running the script saves the calibrated relative viscosity and filtered SYBR images to the '_b.mat' file and displays the calibration:



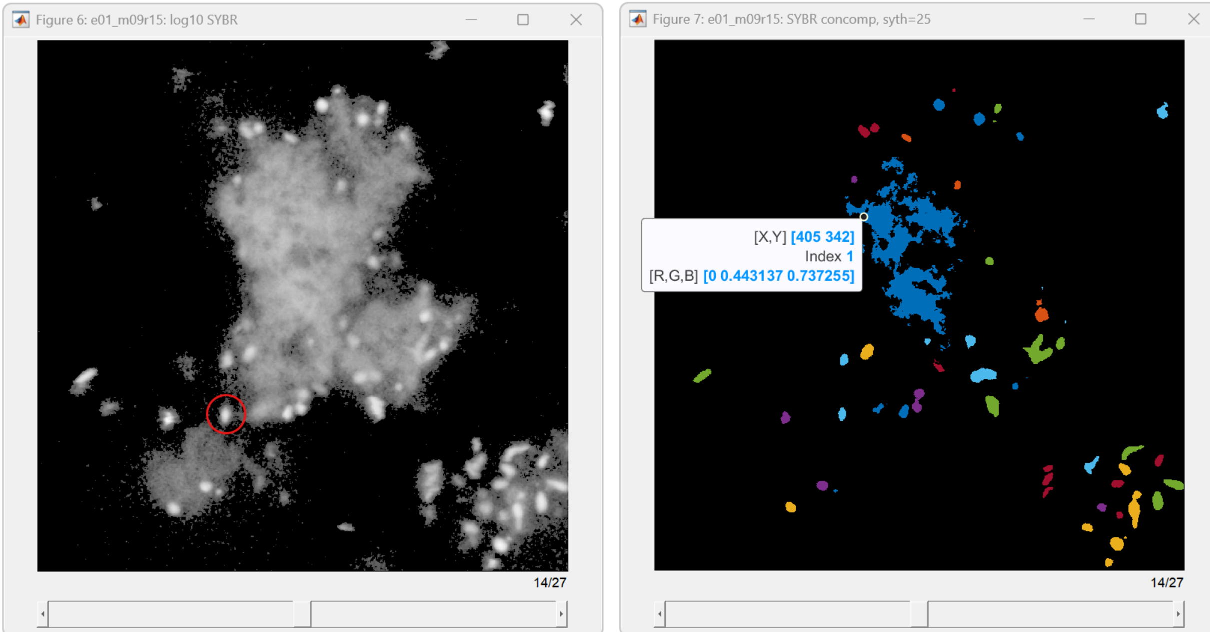
Explanation of the calibration figure panels is provided in the caption of Fig. 1 in the manuscript. The unmixed red RY3, blue RY3, SYBR, and relative viscosity images are also displayed for a plane midway through the z-stack:



To subsequently view all image planes saved in '_a.mat' and '_b.mat' files, use the GUI *viscView.m* with **nm**='e01_m09r15':

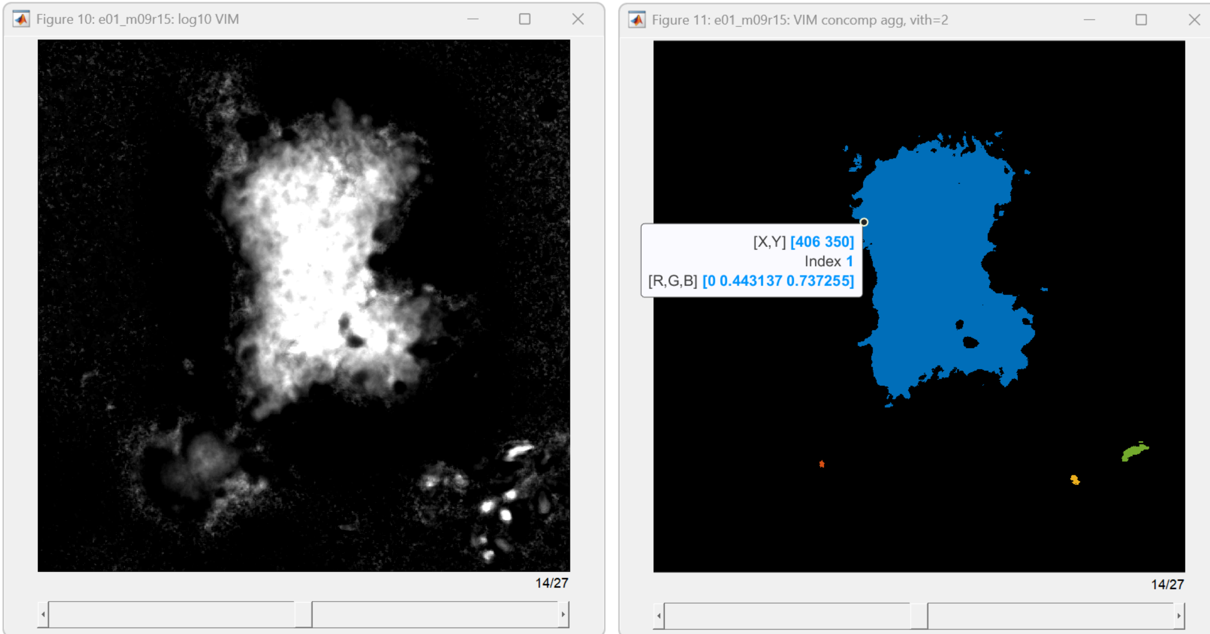


The next step is to determine a SYBR threshold for segmentation of bacterial objects. Because SYBR Green I rapidly photobleaches during excitation, the threshold must be re-evaluated for each aggregate. Edit *rc_c_sybrignore.m* and set **nm**='e01_m09r15' and **syth**=25. Running the script produces two figures, log SYBR on the left and connected components above the **syth** threshold on the right:



Bacteria are brighter, sharply-defined, ~1 um spheres and ellipsoids that may appear as individuals or clusters (e.g. red circle, left panel). Free nucleic acids may be present as larger, lower intensity regions that roughly match the shape of the aggregate. Test higher and lower threshold values to optimize **syth** for (1) including bacteria across all z-stack planes, (2) separating the bacteria signal from any free nucleic acid signal, and (3) maintaining consistent bacterial object sizes across different aggregate images. A histogram of bacterial object volumes is also displayed to help with (3). It is also important at this stage to note the IDs of any non-bacteria objects (free nucleic acids) for use in **syig** in *rc_c_halos.m*. Use the data tip cursor to select a pixel within the object, and the 'Index' displays the object ID (Index 1 in the right figure).

A similar process is used in *rc_c_viscignore.m* to identify viscosity objects to include or exclude from the aggregate analysis in *rc_c_halos.m*. Unlike SYBR Green I, RY3 is highly photostable and the relative viscosity threshold for identifying aggregates should remain the same among all aggregates analyzed. Note that in *rc_c_halos.m* the aggregate is defined by voxels with values greater than **syth** in a connected components analysis and then dilated outward by **edge_buffer** to the final volume. Therefore the edge does not have a constant viscosity value, but should visually conform to the boundary of the aggregate. Edit *rc_c_viscignore.m*, set **nm**='e01_m09r15' and **vith**=2, and run the script:

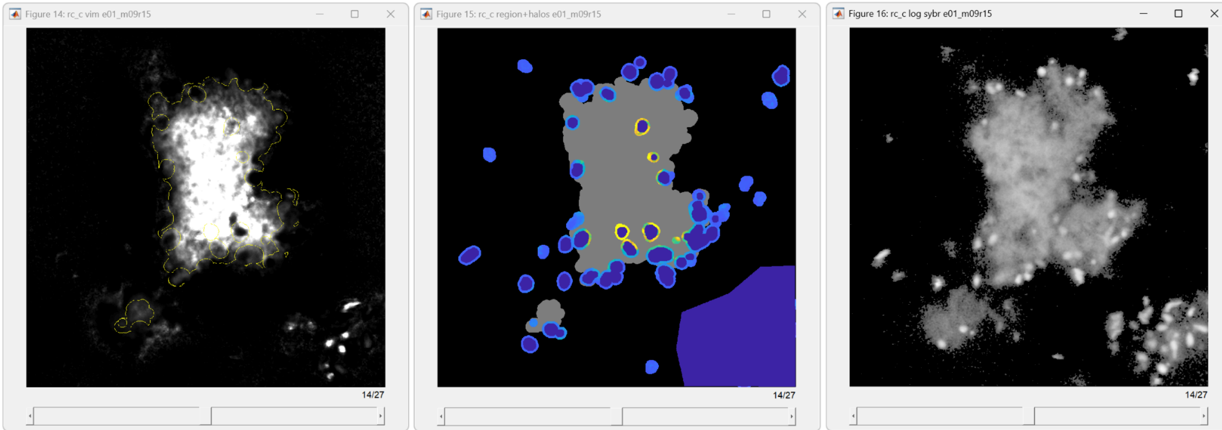


The left panel displays the log of relative viscosity and the right panel shows the connected component viscosity objects. Sometimes bacteria or incomplete aggregates appear as viscosity objects, but should not be identified as aggregates in the analysis. Use the data tip cursor to note the IDs of objects that should be included or excluded from the aggregate analysis (Index 1 in the right figure should be kept in this case).

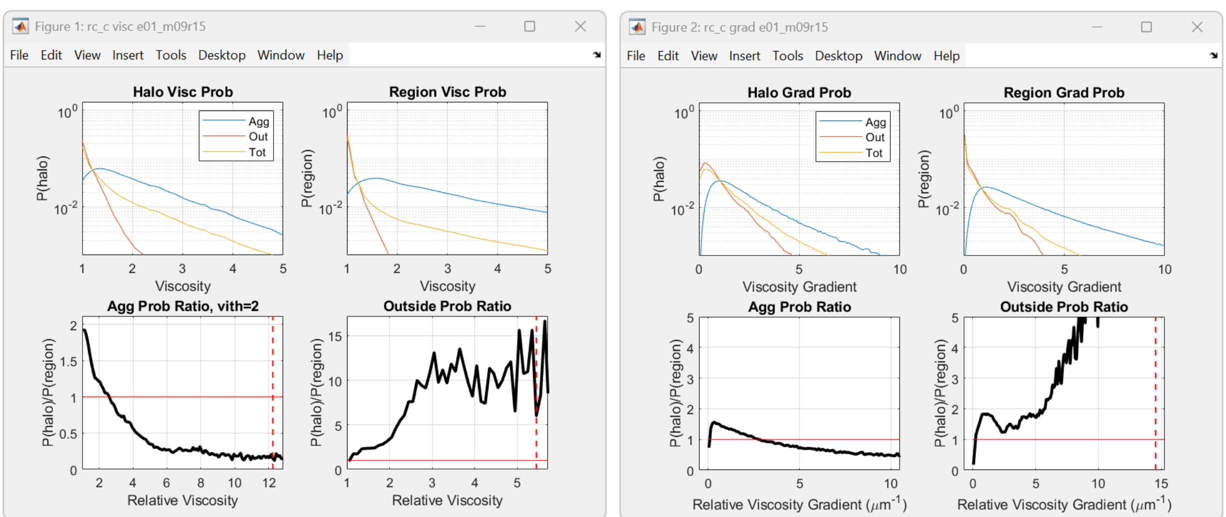
The last step is segmenting the images into bacteria and aggregate volumes and calculating the probabilities of viscosities near bacteria and the aggregates themselves. Edit *rc_c_halos.m* and set **nm**='e01_m09r15'. With **saveout**=true, the segmented images and probabilities will be saved to a '_c.mat' file. These files are fairly large. To reduce loading times and facilitate comparison of probabilities across aggregates, you can also set **svst**='datasetname' to save the probability data to the struct *rcd.mat*. Based on *rc_c_sybrignore.m*, set the SYBR threshold to **syth**=25 and **syig**=[1 2] to not treat free nucleic acids in the aggregate as bacteria. From *rc_c_viscignore.m*, set the viscosity threshold to **vith**=2 and only include two viscosity objects **viig**={'in',[1 2]} as aggregates. You could instead ignore objects with **viig**={'ig',[3:6]}, but cannot use both 'in' and 'ig' at the same time. You can also remove partial aggregates, out-of-focus signal, or moving objects from the analysis using **virm**. In this case, remove a partial aggregate from planes 1-27 and a moving viscosity object from planes 1-13:

```
virm={ 1:14,[682 706 1024 1021 923 833 698],...
            [914 1021 1024 678 681 757 811];
       1:13,[317 319 353 494 584 593],[1 54 117 103 46 1]; };
```

The syntax for viscosity region removal is **virm**={z-planes, column vertices, row vertices}. The vertices can be selected using the data tip cursor while holding shift and clicking sequentially around the object. Right-click on one of the data tips, select 'Export Cursor Data to Workspace…', and enter [cc,cr]=cursorPos(cursor_info) into the command line to retrieve the list of vertices. Running *rc_c_halos.m* saves the results to the files described above and displays the following figures:



Left is the viscosity field with a yellow outline around the aggregates, not including the bacteria. Middle is a view of the aggregate region (gray), outside region (black), and viscosities associated with bacteria (colormap). Both the bacteria and a region removed with **virm** are dark blue. The right panel is log SYBR signal for reference. Note that the bacterial volumes (middle, dark blue) are larger than their corresponding SYBR signals (right panel). This is set by **bact_buffer**=0.4 and prevents any viscosity signal inside the bacteria from being included as aggregate viscosities. The probability data are also displayed:



The left figure shows relative viscosity probabilities and the right shows the probabilities of the magnitudes of the viscosity gradients. In each figure, the top left panel shows the probabilities near the bacteria, separated by region. Top right shows the probabilities for the rest of those regions. The bottom panels show the ratio of the bacterial proximity probabilities to the region probabilities for the aggregate (bottom left) and the outside (bottom right) regions.

Code Files

rc_0_standards.m

Calibrate pairs of viscosity standard images for scaling the experimental images in rc_b_calib.m.

Type: MATLAB script
Requirements: Bio-Formats MATLAB Toolbox (bfmatlab), rc.mat, ryoc.mat, rstd.mat
Input: Location and names of the two raw standard images.
Output: rstd.mat
Parameters:

- Names and files
    - **xs** (*string*): rstd experiment identifier (e.g. 'e17').
    - **xo** (*string*): optical configuration identifier (e.g. 'r15_2' for 2nd set of standards imaged during experiment).
    - **fld** (*string*): directory location of the files, no ending slash.
    - **fns** (*cell*): the raw standard image names, including file extension.
    - **saveout** (*logic*): save the output to rstd.mat?
- Calibration
    - **cmr** (*dbl*): The measured viscosity ratio from the original calibration (e.g. original g30/m01 = 18.7647).
    - **oc450** (*int*): The optical configuration ID number for blue RY3 (I_B) emissions (default: 15).
    - **oc650** (*int*): The optical configuration ID number for red RY3 (I_R) emissions (default: 15).
    - **wv450** (*int*): The raw channel numbers containing blue RY3 (I_B) emissions (default: 3:5).
    - **wv650** (*int*): The raw channel numbers containing red RY3 (I_R) emissions (default: 26).
    - **co** (*int*): Concentrations of RY3 dye to include in the calibration (default: [50 100 200 400 800] μM).
    - **ftp** (*string*): The type of fit used for fitting the adjusted ratio to viscosity (default: 'semi').
    - **pw** (*dbl*): The α tuning parameter (default: 1.1).
    - **um** (*logic*): Use unmixed channels in the calibration (as opposed to only raw)? Default: true.
    - **stn** (*cell*): The identifiers for the viscosity standards, which should match those used in cmr. (e.g. {'m01';'g30'}).
    - **satlvl** (*int*): The maximum image intensity value, indicating a saturated pixel (default: 4095 for a 12 bit image).
- Initialize
    - **ch** (*int*): The indices of channels corresponding to blue RY3 (I_B) emissions in the raw file, red RY3 (I_R) in the unmixed file, and SYBR in the unmixed file. This may vary for other microscopes.

rc_a_deconv.m

Load raw and unmixed experimental image files, extract relevant channels and metadata, and deconvolve with the point spread function of the microscope.

Type: MATLAB script
Requirements: Bio-Formats MATLAB Toolbox (bfmatlab), MATLAB Image Processing Toolbox, psf.mat
Input: Location and names of the raw and unmixed experimental image files.
Output: nm_a.mat
Parameters:

- Files and truncation
    - **nm** (*string*): Experimental image identifier (e.g. ‘e01_m09r15’).
    - **fnu** (*string*): Full path to unmixed experimental image file.
    - **fnr** (*string*): Full path to raw experimental image file.
    - **ztrunc** (*int*): The z-indices of the image stack to save to the output file after deconvolution (e.g. 2:27). Default: [], do not truncate.
    - **psc** (*string*): Path to the psf.mat point spread function data.
    - **plotfig** (*logic*): Plot raw and deconvolved unmixed red RY3 channels?
    - **saveout** (*logic*): Save deconvolved I_B, I_R, SYBR matrices and associated metadata to nm_a.mat?
- Deconvolution
    - **sum1** (*logic*): Force the PSF to sum to 1 before deconvolving?
    - **iter** (*int* in deconvlucy): Number of iterations of the Lucy-Richardson deconvolution (default: 3).
    - **dmpr** (*dbl* in deconvlucy): Dampening threshold (default: 0).
    - **rdot** (*dbl* in deconvlucy): Specify the additive noise (default: 0). Only used if nhood is empty.
    - **nhood** (*int*): Estimate noise in an [m n] pixel neighborhood (default: [3 3]).
    - **satlvl** (*int*): The maximum image intensity value, indicating a saturated pixel (default: 4095 for a 12 bit image).

rc_b_bgselect.m

Select background ROI rectangles in experimental image stacks for use in rc_b_calib.m.

Type: optional MATLAB script
Requirements: roiSelect.m, MATLAB Image Processing Toolbox
Input: nm_a.mat
Output: nm_a.mat
Parameters:

- Files and parameters
    - **nm** (*string*): Experimental image identifier (e.g. ‘e01_m09r15’).
    - **clm** (*dbl* in clim): Color scale limits for visualizing image planes, 'auto' or [cmin cmax].

- **pxth** (*int*): Pixel count goal for background selection in each image plane (default: 200^2).
- **saveout** (*logic*): Save background position metadata to nm_a.mat?

rc_b_calib.m

Filter and calibrate experimental images to relative viscosity.

Type: MATLAB script
Requirements: MATLAB Image Processing Toolbox, rc.mat, ryoc.mat, rstd.mat
Input: nm_a.mat
Output: nm_b.mat
Parameters:

- Files
    - **nm** (*string*): Experimental image identifier (e.g. 'e01_m09r15').
    - **plotfig** (*logic*): Plot calibration and viscosity image figures?
    - **saveout** (*logic*): Save relative viscosity image, filtered SYBR image, and associated metadata to nm_b.mat?
- Image parameters
    - **ubg** (*logic*): Use the image backgrounds to calculate relative viscosity? (Default: true). If false, the viscosity is calculated relative to the lower viscosity standard supplied in rc_0_standards.m (e.g. 'm01' water).
    - **bgin** (*cell*): Set bgin={} to use the backgrounds selected with rc_b_bgselect.m or manually specify the background pixel coordinates as (z indices, row indices, column indices). Example: bgin={1:4, 824:1024, 1:201;...} for a 200 x 200 pixel square in the lower left corner of image planes 1 through 4. Backgrounds must be specified for all z-planes. Note that ubg=false still requires background selection for handling small values and divide-by-zero blowups.
    - **mfwin** (*int*): Median filter pixel window [m n] used for 2D spatial averaging of unmixed red RY3 (I_R) and SYBR image planes (default: [5 5]).
    - **gawin** (*dbl*): The standard deviation of the Gaussian filter smoothing kernel used for 2D spatial averaging of unmixed blue RY3 (I_B) image planes (default: 5).
- Calibration parameters (should match rc_0_standards.m for a given optical configuration)
    - **oc450** (*int*): The optical configuration ID number for blue RY3 (I_B) emissions (default: 15).
    - **oc650** (*int*): The optical configuration ID number for red RY3 (I_R) emissions (default: 15).
    - **wv450** (*int*): The raw channel numbers containing blue RY3 (I_B) emissions (default: 3:5).
    - **wv650** (*int*): The raw channel numbers containing red RY3 (I_R) emissions (default: 26).
    - **co** (*int*): Concentrations of RY3 dye to include in the calibration (default: [50 100 200 400 800] µM).
    - **ftp** (*string*): The type of fit used for fitting the adjusted ratio to viscosity (default: 'semi').
    - **pw** (*dbl*): The α tuning parameter (default: 1.1).

- **um** (*logic*): Use unmixed channels in the calibration (as opposed to only raw)? Default: true.
- **xs** (*string*): Optional experiment identifier for rstd.mat (default: ‘’, uses first three characters of nm).
- **xg** (*string*): Optional postfix for the optical configuration identifier in rstd.mat (default: ‘’). Example: when viscosity standards were imaged multiple times during an experiment, xg=’_2’ and oc650=15 corresponds to xo=‘r15_2’ set in rc_0_standards.m.

rc_c_sybrignore.m

View the SYBR ‘b’ images next to thresholded images with labelled connected components (bacteria and other cell objects). This is helpful for determining what threshold value to use for identifying cells with SYBR in rc_c_halos.m. Use the figure data cursor to determine the ID of an object to be manually included or excluded from analysis in rc_c_halos.m.

Type: optional MATLAB script
Requirements: MATLAB Image Processing Toolbox
Input: nm_b.mat
Output: none
Parameters:
- Files
  - **nm** (*string*): Experimental image identifier (e.g. ‘e01_m09r15’).
- Analysis parameters
  - **syth** (*dbl*): Lower threshold value for detecting SYBR signal, indicating bacteria or other cells.
  - **bact_rad** (*dbl*): The expected radius (μm) of the SYBR signal produced by the smallest spherical bacteria present in the sample (default: 0.35). This is used for differentiating bacteria from noise. Note that the nucleic acid signal (SYBR) inside a bacterial cell is smaller than the cell radius.

rc_c_viscignore.m

View the viscosity ‘b’ images next to thresholded images with labelled connected components (mucus objects). Use the figure data cursor to determine the ID of an object to be manually included or excluded from analysis in rc_c_halos.m.

Type: optional MATLAB script
Requirements: MATLAB Image Processing Toolbox
Input: nm_b.mat
Output: none
Parameters:
- Files
  - **nm** (*string*): Experimental image identifier (e.g. ‘e01_m09r15’).
- Analysis parameters

- **vith** (*dbl*): Lower threshold value for determining the edge viscosity of the mucus region.
- **bact_rad** (*dbl*): The expected radius (µm) of the SYBR signal produced by the smallest spherical bacteria present in the sample (default: 0.35). This is used for differentiating bacteria from noise. Note that the nucleic acid signal (SYBR) inside a bacterial cell is smaller than the cell radius.

rc_c_halos.m

Identify bacteria or other cell types and draw 'halo' regions around them; partition mucus viscosity into edge and inner regions that do not include bacteria or halos; calculate the probability of a viscosity occurring in halos and regions for the edge, inner, and outer regions. The nm_cv.m output contains viscosity data, and the nm_cg.m output contains viscosity gradient data.

Type: MATLAB script
Requirements: Binary image distance transform for variable aspect ratio (bwdistsc), MATLAB Image Processing Toolbox
Input: nm_b.m
Output: nm_cv.m, nm_cg.m
Parameters:

- Files
  - **nm** (*string*): Experimental image identifier (e.g. 'e01_m09r15').
  - **plotfig** (*logic*): Plot viscosity probability and region visualization figures?
  - **saveout** (*logic*): Save viscosity probability, separated viscosity region matrices, and associated metadata to nm_b.mat?
  - **svst** (*string*): Dataset identifier for saving viscosity probabilities to rcd.mat (default: '', do not save).
  - **xg** (*string*): Optional tag for the output filename for testing different parameters without overwriting existing files (default: ''). Example: e01_m09r15_2_cv.mat.
- Analysis parameters
  - **syth** (*dbl*): Lower threshold value for detecting SYBR signal, indicating bacteria or other cells.
  - **vith** (*dbl*): Lower threshold value for determining the edge viscosity of the mucus region.
  - **bact_rad** (*dbl*): The expected radius (µm) of the SYBR signal produced by the smallest spherical bacteria present in the sample. This is used for differentiating bacteria from noise. Note that the nucleic acid signal (SYBR) inside a bacterial cell is smaller than the cell radius.
  - **syig** (*int*): Array of SYBR connected component IDs (possible bacterial objects) to ignore in the analysis (default: [], none). Determined using rc_c_sybrignore.m. Example: [2 5:7].
  - **viig** (*cell*): Cell array of viscosity connected component IDs (mucus objects) to ignore or include in the analysis (default: {}, none). Syntax: 'ig'=ignore, 'in'=include only for 1=inner, 2=edge regions. Example: { 'ig1',[2 7]; 'in2',1; } ignores inner region IDs 2 and 7 and *only* includes edge region ID 1 (ignoring all other edge region objects).
  - **virm** (*cell*): Remove viscosity image volumes specified by polygons with coordinates (z indices, column vertices, row vertices) from the analysis. Default: {}, none. Example: {

3:5,[1 50 1],[1 1 60]; } removes a triangular prism from z-planes 3 through 5. Multiple overlapping volumes are allowed.
    - **vike** (*cell*): Keep a viscosity image volume specified by polygons with coordinates (z indices, column vertices, row vertices) and remove the remaining volume from analysis. Default: {}, none. Example: { 3:5,[1 50 1],[1 1 60]; } removes everything except a triangular prism from z-planes 3 through 5.
- Initialize
    - **bact_rad** (*dbl*): The expected radius (µm) of the SYBR signal produced by the smallest spherical bacteria present in the sample (default: 0.35). This is used for differentiating bacteria from noise. Note that the nucleic acid signal (SYBR) inside a bacterial cell is smaller than the cell radius.
    - **bact_buffer** (*dbl*): The distance (µm) between the SYBR signal and the inner edge of the bacterial halo (default: 0.4). This includes both a portion of the bacterial cell (which is larger than the SYBR signal) and an additional buffer to ensure that viscosity signal inside the bacterial cell does not overlap with the halo volume.
    - **halo_delta** (*dbl*): The thickness (µm) of the halo surrounding bacteria, in which bacterial associations with mucus viscosity are calculated (default: 0.3).
    - **edg_buffer** (*dbl*): The distance (µm) to extrude the edge viscosity region defined by the vith threshold (default: 0.8).
    - **visc_max** (*dbl*): The upper bin edge for calculating the probability of relative viscosities.
    - **visc_delta** (*dbl*): The bin width for relative viscosity probabilities.
    - **grad_max** (*dbl*): The upper bin edge for calculating the probability of the magnitude of the relative viscosity gradient.
    - **grad_delta** (*dbl*): The bin width for the magnitude of relative viscosity gradient probabilities.

roiSelect.m

Creates a GUI for selection of rectangle ROIs in a z-stack and returns ROI Positions for each z-plane in a cell array.

Type: MATLAB function used by rc_b_bgselect.m.
Requirements: MATLAB Image Processing Toolbox
Output: roip, a size(img,3) x 1 cell array
Description:
roip = roiSelect creates a GUI for selection of rectangle ROIs in a default z-stack and returns ROI Positions for each z-plane in a cell array. The image color scale is log10. There is no check for overlapping ROIs.

roip = roiSelect(img) uses the specified 2-D or 3-D image.

roip = roiSelect(img,clm) also defines the color scale limits as 'auto' (default) or [cmin cmax].

roip = roiSelect(img,clm,pixelThresh) also defines the pixel count goal (default: 200^2).

roip = roiSelect(img,clm,pixelThresh,roiPos) also pre-populates the ROIs using roiPos, a size(img,3) x 1 cell array.

viscView.m

A GUI used to simultaneously view individual planes of the red RY3, blue RY3, SYBR, and relative viscosity images saved in the '_a.mat' and '_b.mat' files.

Type: MATLAB script
Requirements: MATLAB Image Processing Toolbox
Input: nm_a.m, nm_b.m
Output: none
Parameters:
- Files
    - **nm** (*string*): Experimental image identifier (e.g. 'e01_m09r15').
    - **clm** (*dbl* in clim): Color scale limits for visualizing the relative viscosity, 'auto' or [cmin cmax].

cursorPos.m

Function to retrieve column and row positions from exported data tip cursor data.

Type: MATLAB function
Requirements: none
Output: cc, cr
Description:
[cc,cr]=CURSORPOS(cursorInfo) cursorInfo is the exported cursor data variable, image position is (cc = X cols, cr = Y rows).

Data Files

rc.mat

Contains the struct 'r' with raw and unmixed emission data from varying dye concentrations and viscosity standards.

ryoc.mat

Contains the cell array 'ryoc' that defines the optical configurations (OCs) used in calibration. Each row number is used as the OC identifier rZZ. The columns are the full name, laser power, gain, line averaging, and pixel dwell time parameters.

rstd.mat

Contains the struct 'rstd' with calibrated viscosity standards for each experiment. Structure:
- **rstd**

- **eXX**: The experiment identifier (e.g. e17)
    - **rZZ**: The optical configuration identifier (e.g. r15)
        - **mYY**: The calibrated viscosity of the lower viscosity standard (e.g. m01, water)
        - **gYY**: The calibrated viscosity of the higher viscosity standard (e.g. g30, viscosity 30 aqueous glycerol)
        - **gmr**: The experimental ratio gYY/mYY.
        - **cmr**: The original calibration ratio gYY/mYY.

psf.mat

Contains the point spread function of the microscope as a 3D matrix and associated metadata.